

Fleet Maintenance Optimization:

A Comprehensive Cost & Event Analysis (Jan 2023 – June 2024)

Introduction

When I first opened this dataset of over 25,0000 vehicle service logs, my mind went blank. I honestly thought, *“What am I even going to find here?”* The data seemed too limited at first glance. There weren't many models, and without vehicle registration numbers, I couldn't even pinpoint exactly which individual vehicle was burning the most cash.

But then it clicked.

I noticed the date segment, meaning I could track exactly which months spiked in costs. Then I saw the other variables: road types, climate conditions, maintenance types, and service levels. Suddenly, my vision expanded. I realized I was looking at a massive combination of facts that could tell a hundred different stories.

So, welcome to my fleet project. This isn't a flashy, entertaining data story. It's just the raw reality of a 260M fleet, broken down into its true 6-month chapters. Let's dive in.

Project Toolkit

To process over 25,000 service logs and uncover these patterns, I built an end-to-end data pipeline using this modern analytics stack:

- BigQuery & SQL: Heavy-duty data warehousing and root-level view optimization to query millions in fleet costs instantly.
- Excel & Power Query: Initial data cleaning, transformation, and structural shaping before injection into the warehouse.
- Power BI & DAX: Advanced time-intelligence modeling, interactive cross-filtering, and executive-level KPI dashboards.
- GitHub: Version control and public repository hosting for project deployment transparency.

Project Assets

- Live Repository: [GitHub Project Space](#)
- RawDataset: [Fleet_Managment_Project/FLEET_MANAGMENT_PROJECT/MAIN_DATA_SOURCE](#) at main · abhishek862000/Fleet_Managment_Project · GitHub

Page 1: Vehicle details and maintenance reports

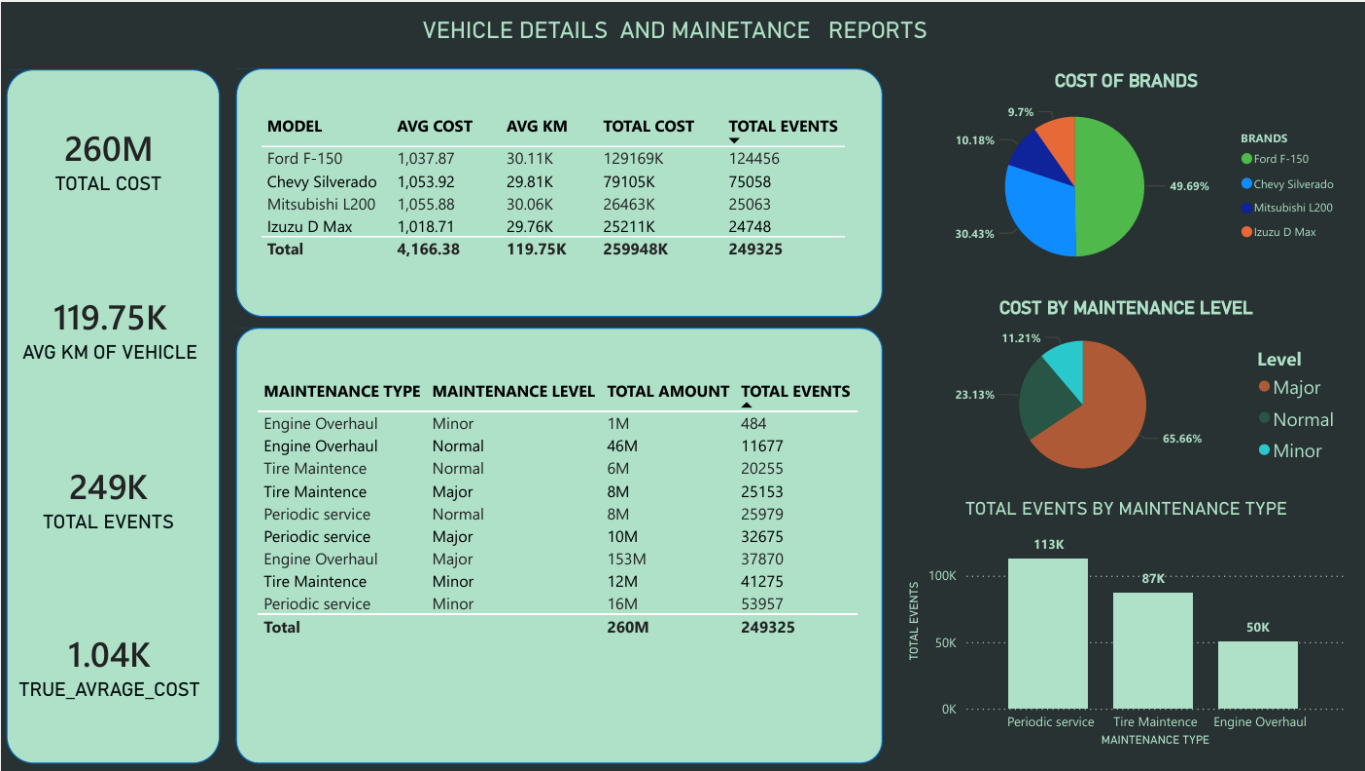


Fig 1.1

The 260M Reality Check

Looking at the main dashboard view in , the high-level totals are massive:

- Total Cost: 260M
- Total Events: 249K maintenance check-ins
- True Average Cost: 1.04K per single event

Here is the breakdown by the numbers so it is easy to catch.

1. The Vehicle Models (Who is spending?)

- Ford F-150 (The King of Expenses):
 - 129M total cost (almost half the entire budget)
 - 124K total events
- Chevy Silverado:
 - 79M total cost
 - 75K total events
- Mitsubishi L200 (Highest Avg Cost):
 - 26M total cost
 - 25K total events
 - 1,055.88 highest average cost per service
- Isuzu D-Max:
 - 25M total cost
 - 25K total events

The Data Blindspot: The tables in Fig 1.1 group everything cleanly by brand, but they hide specific vehicle IDs or registration numbers. We can see *which model* burns money, but we

cannot pinpoint the exact rogue trucks destroying the budget. Future data sampling is required to fix this.

2. Maintenance Types (What are we paying for?)

- Engine Overhauls (The Cash Killer):
 - 200M total cost (swallows the vast majority of the budget)
 - 50K events (*lowest event count, but highest cost*)
- Periodic Services:
 - 34M total cost
 - 113K events (*highest event count, but cheap*)
- Tyre Maintenance:
 - 26M total cost
 - 87K events

3. Maintenance Intensity (How severe is it?)

- Major Maintenance: 171M total cost | 96K events
- Normal Maintenance: 60M total cost | 58K events
- Minor Maintenance: 29M total cost | 96K events

The Bottom Line

Almost half our money goes directly to Ford F-150s, and 200M of our total 260M cash is completely swallowed up by engine overhauls. We know *what* is costing us money; our next step is using the monthly interactive features to track *when* and *where* to stop the bleeding.

Page 2 : 2023 Jan - 2024 June cost analysis by month

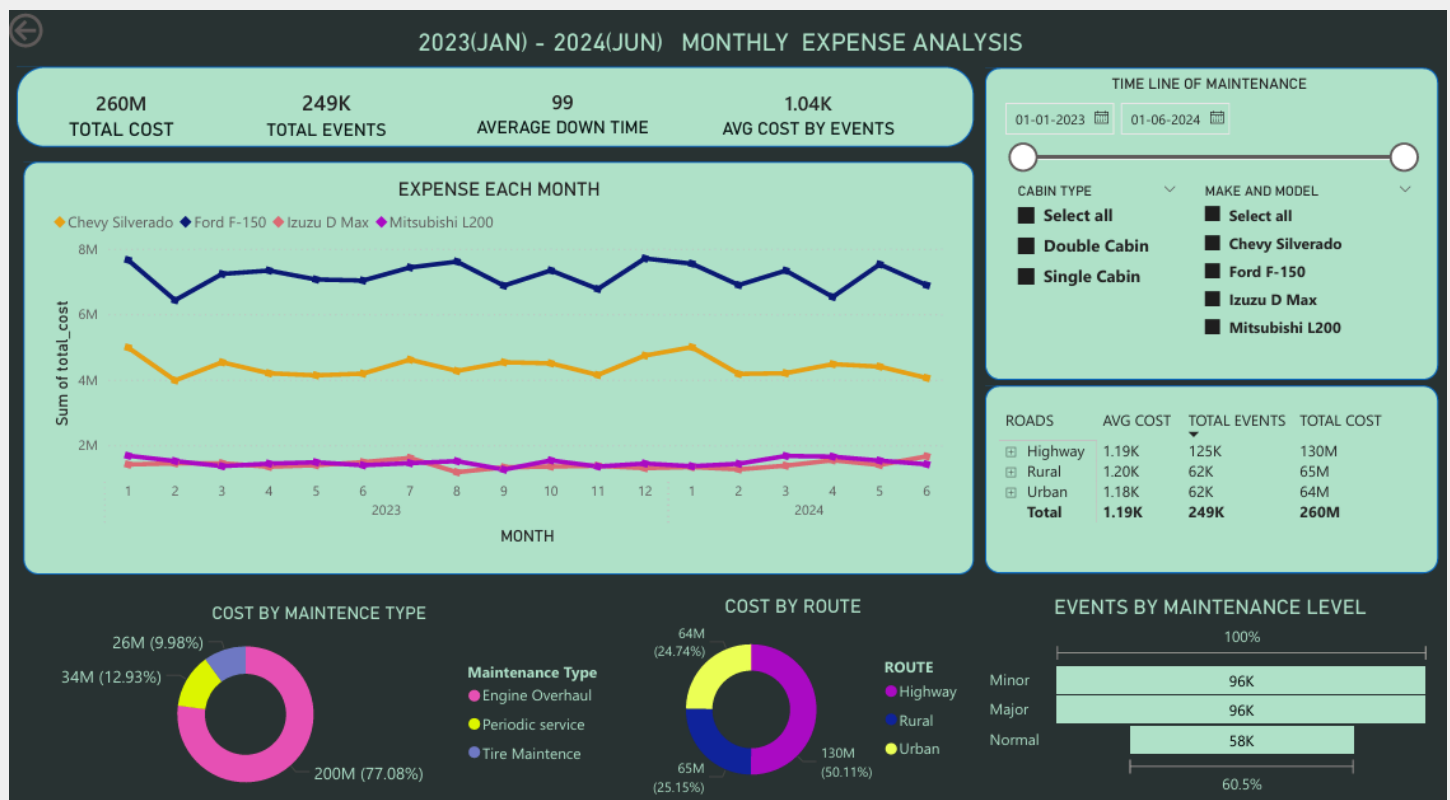


Fig 2.1

Page 2: The Monthly Breakdown

This page is built for interactive filtering. As shown in image_c1e481.png, you can use the timeline slider, select cabin types, or filter by routes to instantly update the KPIs and charts. Here are the hard facts, broken down by the numbers.

1. Double Cabin vs. Single Cabin (Body Type Performance)

- Double Cabins:
 - 144M total cost
 - 138K events
 - 100 hours average downtime
 - 1.04K average cost per service
- Single Cabins:
 - 116M total cost
 - 112K events
 - 99 hours average downtime
 - 1.04K average cost per service

2. The Monthly Burn (Steady Brand Spending)

The monthly line graph trends remain identical every single month:

- Ford F-150: 7M – 8M per month (*Always highest*)
- Chevy Silverado: 4M – 5M per month (*Steady second*)
- Isuzu D-Max & Mitsubishi L200: 1M – 2M per month (*Flat at the bottom*)

3. The 6-Month Copy-Paste (Timeline Intervels)

The fleet behaves almost identically across every 6-month chapter:

- Jan–Jun 2023: 86M total cost | 82K events | 43M highway cost

- Jul–Dec 2023: 87M total cost | 83K events | 43M highway cost
- Jan–Jun 2024: 87M total cost | 84K events | 43M highway cost
- (*Baseline shown in i fig 2.1*)

4. Quick Operational Facts

- Route Costs:
 - Highway: 130M total cost
 - Rural: 65M total cost
 - Urban: 64M total cost
- Climate Costs:
 - Clear: 69M (79.09%)
 - Rainy: 14M (15.62%)
 - Snowy: 8 M (3.1%)
 - Windy: 5 M (2%)

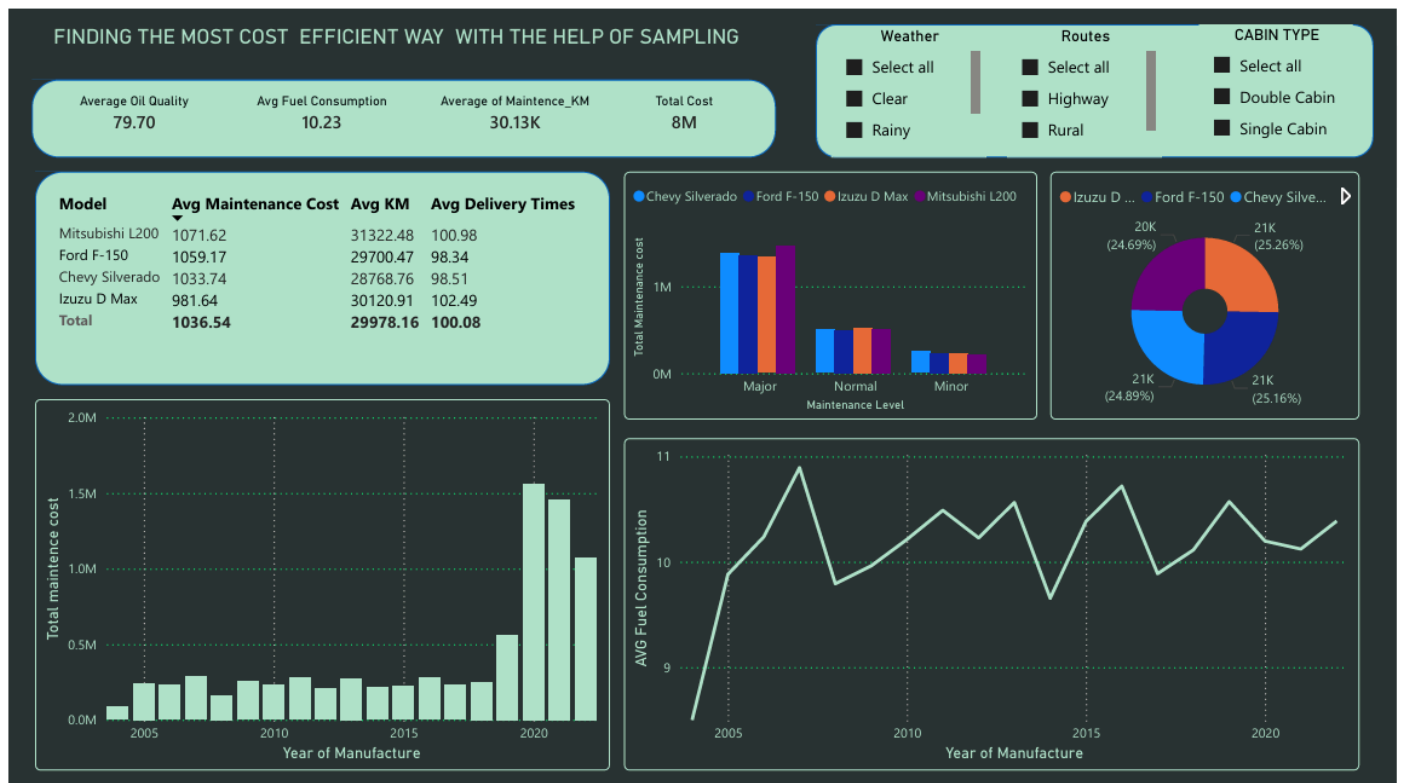
Page 3 :Finding the most cost efficient models and conditions with the help of sampling

What we did here ?

Using a CTE named `RankedData`, we implemented stratified random sampling across 249,000 records. By partitioning by `Make_and_Model` and `Vehicle_Type` and sorting randomly via `RAND()`, the query filters for `row_num <= 1000` to draw an even sample of 1,000 events per machine/cabin category for custom modeling

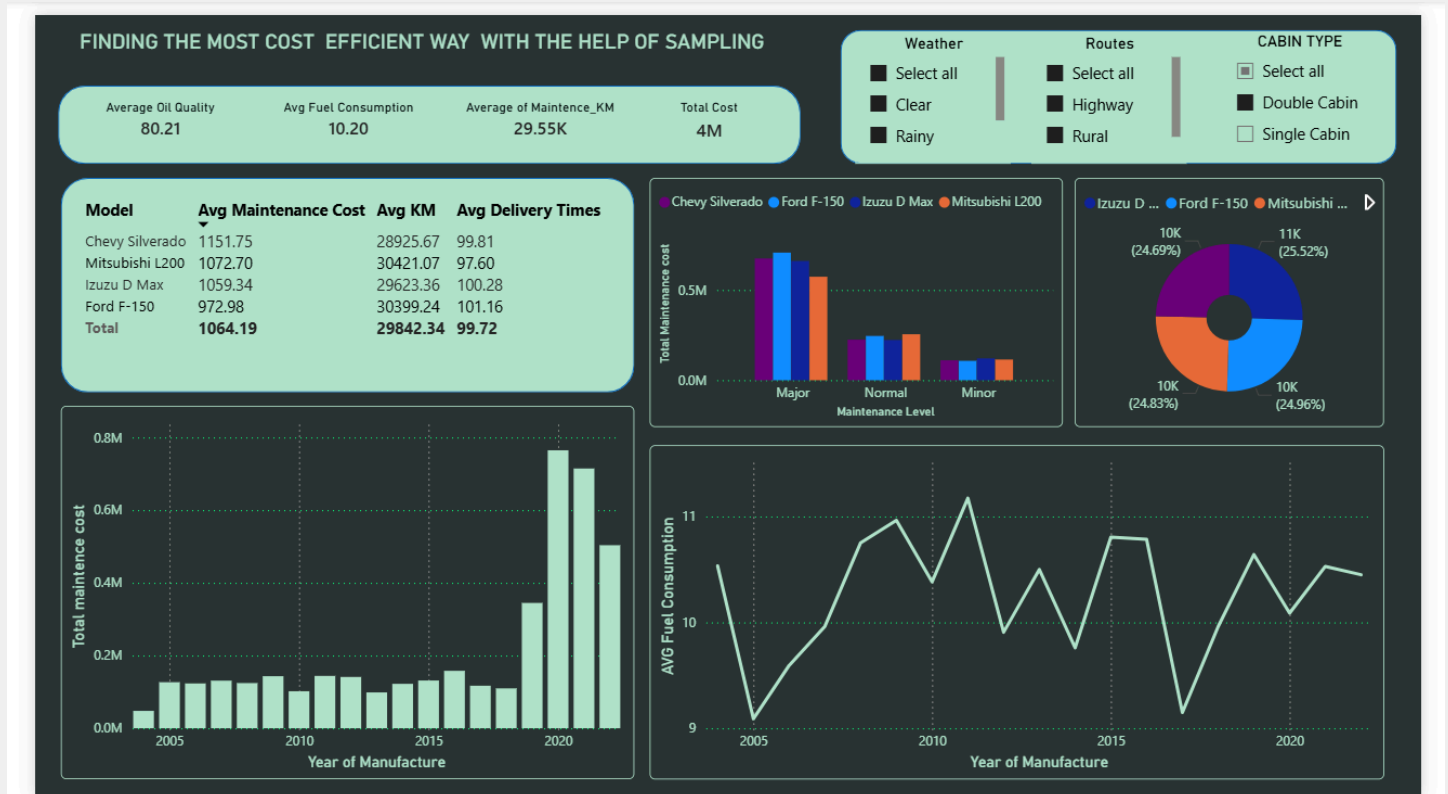
```
WITH RankedData As( -- ranked data is the new temprry table viw
SELECT *,
    row_number() over(
        partition by Make_and_Model,Vehicle_Type -- for cereating the partition
        order by rand() -- ordering randomily
    )row_num --refed as row number like method in a normal programin
FROM `fleetmaintenance-862000.Fleet.fleet`
)
select * from RankedData where row_num<=1000; --select all the data from the ranked data and called the method
```

Page 1: Fleet Baseline & Initial Findings



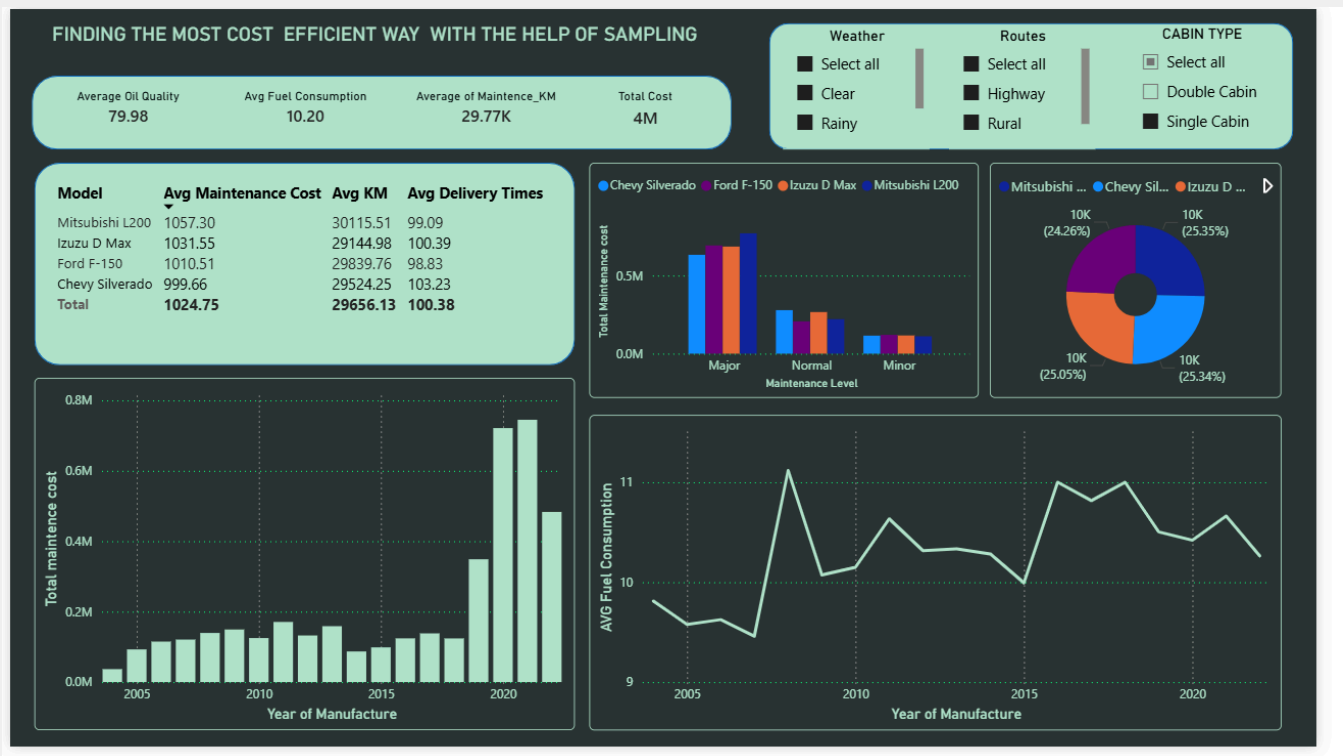
- **Overall Maintenance Trend:** Shows total maintenance costs by year of manufacture (2005–2022). Older models maintain a low, flat maintenance cost, but there is a dramatic spike in total costs for models built in 2020 and later (peaking around \$1.5M+ per model year).
- **Baseline Fleet Table:** Evaluates the overall average performance across all 4 truck models:
 - Mitsubishi L200: Highest overall maintenance cost (\$1,071.62) and highest average distance (31,322 KM).
 - Ford F-150: \$1,059.17 avg cost / 29,700 KM.
 - Chevy Silverado: \$1,033.74 avg cost / 28,768 KM.
 - Isuzu D-Max: Lowest overall maintenance cost (\$981.64) / 30,120 KM.
 - Fleet Totals/Averages: \$1,036.54 avg maintenance cost, 29,978 KM avg distance, and 100.08 avg delivery time.
- **General Metrics:** Average Oil Quality is 79.70, and Average Fuel Consumption is 10.23.

Page 2: Fuel Consumption Analysis & Double Cabin Filter



- **Fuel Efficiency Over Time:** Displays average fuel consumption from 2005 to 2022, fluctuating around 10.0 to 11.0 L/100km.
- **Double Cabin Metrics:** Filtering specifically for Double Cabin vehicles changes the model cost rankings:
 - Chevy Silverado: Highest maintenance cost (\$1,061.07).
 - Ford F-150: \$1,003.35 avg maintenance cost.
 - Isuzu D-Max: \$995.03 avg maintenance cost.
 - Mitsubishi L200: Becomes the most cost-efficient model (\$938.72).
- **Double Cabin Averages:** Average maintenance cost drops to \$999.04, average distance is 29,053 KM, and fuel consumption sits at 10.41.
- **Maintenance Levels:** Shows that major maintenance makes up the largest proportion of total costs compared to normal and minor repairs.

Page 3: Single Cabin Filter & Cost Inversion



- **Single Cabin Metrics:** Filtering strictly for Single Cabin vehicles flips the model rankings:
 - Mitsubishi L200: Becomes the most expensive model (\$1,057.30).
 - Isuzu D-Max: \$1,031.55 avg maintenance cost.
 - Ford F-150: \$1,010.51 avg maintenance cost.
 - Chevy Silverado: Becomes the most cost-efficient model (\$999.66).
- **Single Cabin Averages:** Overall average maintenance cost rises to \$1,024.75, average distance is 29,656 KM, and fuel consumption averages 10.20.
- **Cost Share Distribution:** Shows an almost equal ~25% share of maintenance workload across all four models when set to single cabin setups.

Summary Comparison Table

Page / Filter	Most Cost-Effective Model	Highest Cost Model	Avg Maintenance Cost	Avg Distance (KM)
Page 1 (Overall)	Izuzu D-Max (\$981.64)	Mitsubishi L200 (\$1,071.62)	\$1,036.54	29,978 KM
Page 2 (Double Cabin)	Mitsubishi L200 (\$938.72)	Chevy Silverado (\$1,061.07)	\$999.04	29,053 KM
Page 3 (Single Cabin)	Chevy Silverado (\$999.66)	Mitsubishi L200 (\$1,057.30)	\$1,024.75	29,656 KM

Conclusion

This analysis proves that while high-level totals highlight the biggest cash sinks, targeted sampling and granular cross-filtering reveal where operational efficiency truly lies. Over the 18-month analysis period (Jan 2023 – June 2024), three primary insights emerge to guide future fleet strategy:

1. Targeted Fleet Allocation by Cabin Type: There is no single "cheapest" vehicle model across all operations. The Mitsubishi L200 delivers maximum cost efficiency when configured as a Double Cabin (\$938.72 avg maintenance cost), whereas the Chevy Silverado leads efficiency in Single Cabin configurations (\$999.66 avg maintenance cost). Purchasing and deployment decisions should strictly pair model choices with cabin requirements.
2. Addressing the Engine Overhaul Drain: With 200M of the total 260M budget swallowed up by engine overhauls alone, preventative maintenance strategies must be restructured. Shifting operational focus toward proactive periodic servicing—which currently accounts for 113K events at a fraction of major repair costs—will directly curtail expensive major component failures.
3. Closing the Vehicle ID Blindspot: While macro trends clearly identify the Ford F-150 as absorbing nearly half of total expenditure (129M), optimizing the fleet further requires tracking individual vehicle registration numbers. Implementing unique vehicle identifiers in future data logging will allow fleet managers to isolate outlier rogue assets and evaluate repair-vs-replace thresholds accurately.